



Relation-Guided and Patient-Adaptive Hypernetworks for Multi-Task Assessment in Corneal Confocal Microscopy

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ABSTRACT

Corneal confocal microscopy (CCM) provides a non-invasive window into ocular and systemic neuroimmune alterations, but clinical translation is hindered by manual interpretation and fragmented workflows. Existing deep-learning methods focus on isolated tasks, providing limited support for unified patient-level assessment. Furthermore, unified assessment faces challenges like cross-center variability, visually ambiguous phenotypes, and the need for patient-specific outcome mappings. To address these challenges, we propose a unified prior-guided and patient-adaptive framework for CCM imagery assessment. It comprises: (1) relation-guided task-specific feature extraction to retain clinically robust visual factors; (2) patient-semantic-guided feature contextualization to resolve visually ambiguous phenotypes; and (3) patient-task-conditioned hypernetworks that adapt predictive mappings to specific patient states and task identities. Together, these components unify ocular and systemic neuropathy diagnosis, continuous metric estimation, and both short- and long-term prognoses. Across cohorts comprising 35,039 CCM images from eight centers, the framework achieved state-of-the-art performance across six clinical objectives and maintained robustness under cross-center shifts. Beyond predictive accuracy, the estimated metrics preserved established clinical relationships with dry eye disease severity, short-term outcomes, and diabetic peripheral neuropathy status. These findings highlight the potential of CCM-derived digital biomarkers for integrated disease phenotyping. The source code is publicly available at <https://github.com/LHS-A/CCM-DiagProg>.

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1. Introduction

The cornea serves as an accessible window into neuroimmune health, with corneal confocal microscopy (CCM) enabling non-invasive, cellular-resolution visualization of the sub-basal nerve plexus and resident immune cells (Gad et al., 2022;

Alam et al., 2022). This microscopic phenotyping provides the biological evidence necessary to stratify disorders that share overlapping symptoms but are driven by distinct mechanisms. As visualized in Fig. 1, healthy corneas exhibit a dense, uniform nerve network. In contrast, dry eye disease (DED) is driven by inflammatory activation, marked by the prominent infiltration of immune cells (Langerhans cells). Neuropathic corneal pain (NCP) presents a structurally decoupled phenotype defined by localized neural aberrations, such as microneuromas, stemming from somatosensory dysfunction (Dieckmann et al., 2017; Moein et al., 2020). Beyond localized ocular conditions, CCM also non-invasively tracks systemic neuropathies such as diabetic peripheral neuropathy (DPN). Here, the transition from Type 1 diabetes mellitus without diabetic peripheral neuropathy (DPN-) to Type 1 diabetes mellitus with diabetic peripheral neuropathy (DPN+) is characterized by profound, progressive corneal nerve loss rather than isolated immune or neuroma events. When interpreted jointly with metabolic markers like HbA1c, these distinct structural phenotypes reliably stratify neuro-metabolic risk (Braffett et al., 2020).

To provide comprehensive patient management, however, these microscopic phenotypes must be translated beyond disease identification into quantitative clinical assessment and future outcome prediction. For ocular surface diseases, structural alterations are associated with functional endpoints such as tear-film break-up time (TBUT), corneal fluorescein staining (CFS), tear secretion (SIT), and patient-reported symptoms (OSDI) (Wolffsohn et al., 2017). Their longitudinal evolution further reflects treatment response and postoperative recovery, motivating the use of baseline CCM for both one-month treatment-response prediction and longer-term prognosis. For systemic neuropathy, joint interpretation with metabolic markers such as HbA1c provides complementary information for neuro-metabolic risk assessment (Braffett et al., 2020). Despite these biological links, current assessment remains fragmented: CCM interpretation is largely manual (Alam et al., 2022), while complementary functional and metabolic measurements require separate contact-based, dye-assisted, questionnaire-based, or blood-based procedures. A unified framework that maps CCM phenotypes to diagnosis, continuous clinical metrics, and longitudinal outcomes could therefore provide a more integrated patient-level assessment.

Recent deep-learning approaches for CCM have predominantly targeted isolated structural segmentation, e.g., corneal nerve and Langerhans cell, or end-to-end disease classification (Li et al., 2025b,c,a; Mou et al., 2022; Zhao et al., 2020; Koseoglu et al., 2025; Preston et al., 2022; Chen et al., 2024). While demonstrating technical feasibility, these methods remain largely confined to isolated anatomical structures or individual clinical objectives, limiting unified patient-level assessment. To our best knowledge, no prior automated CCM study has directly regressed clinically established continuous endpoints (CFS, TBUT, SIT, OSDI, or HbA1c). Furthermore, no framework integrates complex DED, NCP and DPN stratification, continuous metric regression, and longitudinal prognostic assessment (both short-term treatment and long-term postoperative) within a single framework. Consequently, current sys-

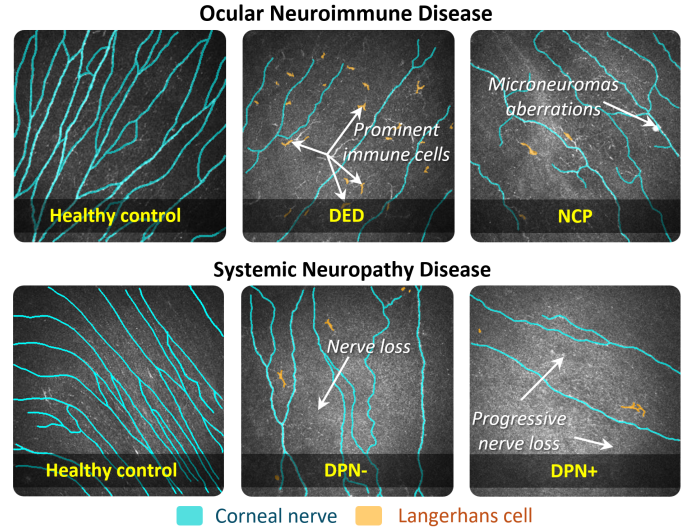


Fig. 1. Distinct microscopic imaging characteristics of ocular neuroimmune and systemic neuropathic diseases via CCM.

tems remain fragmented across isolated tasks and provide limited support for unified patient-level assessment spanning disease discrimination, quantitative clinical characterization, and longitudinal monitoring.

Developing this much-needed unified framework introduces two critical computational bottlenecks. First, *task conflicts and acquisition/cohort shifts destabilize learned features*. Categorical risk stratification, continuous regression, and prognosis necessitate heterogeneous visual evidence. Stratification requires invariant structural boundaries, while regression demands high sensitivity to localized morphological changes. Forcing these divergent objectives into a shared feature space induces gradient conflicts and negative transfer. Unmitigated acquisition variability and cross-center cohort shifts exacerbate this instability, obscuring mechanism-grounded neuroimmune markers behind imaging nuisance associations. Second, *rigid task heads fail to resolve patient-specific functional mappings*. Conventional multi-head or Mixture-of-Experts (MoE) architectures restrict sample-dependent routing to a finite set of fixed configurations, failing to dynamically adapt the prediction function to the patient's unique clinical state and objective. Overcoming these barriers requires a robust architecture capable of structurally decoupling visual evidence to resolve task conflicts, contextualizing this evidence with the patient's semantic records, and dynamically generating customized prediction parameters.

To address these clinical and computational challenges, we propose a unified, patient-adaptive framework for CCM-based ocular neuroimmune and systemic neuropathy assessment. Our approach spans early warning risk stratification, continuous clinical metric regression, and longitudinal prognosis by progressively resolving heterogeneity across features, patient contexts, and prediction functions. First, relation-guided task-specific feature extraction organizes latent visual dependencies using clinically induced relations, and applies HSIC/KCI-based coarse-to-fine filtering. This allows each objective to retain its own target-relevant and conditionally robust visual factors rather than relying on a rigid shared representation.

Second, patient-semantic-guided feature contextualization integrates patient-specific clinical data to reinterpret visual evidence, thereby complementing visually similar CCM phenotypes with patient-specific clinical context while preserving an image-only pathway when such information is incomplete. Third, a patient-task-conditioned hypernetwork dynamically generates custom prediction parameters for every patient-task pair. By replacing static population-level task heads with dynamically conditioned functions, the framework adapts to both heterogeneous clinical objectives and individual patient states. This enables the final mapping to adapt not only across heterogeneous objectives but also across patient states within the same objective.

The main contributions of this work are threefold:

- To our knowledge, this is the first automated CCM framework to jointly execute neuroimmune and systemic neuropathy diagnosis, alongside continuous functional metric regression, short-term treatment and long-term postoperative prognosis. By structurally decoupling multi-view features, it achieves precise stratification of healthy controls, DED, NCP, and DPN.
- The framework learns patient-adaptive mappings to directly predict established macroscopic functional outcomes (CFS, TBUT, SIT, OSDI, HbA1c) from microscopic CCM phenotypes. This reduces reliance on fragmented, contact-based measurements and enables objective neuro-metabolic risk tracking.
- We introduce a dynamic architecture that moves beyond fixed feature sharing. By combining HSIC/KCI-based feature filtering, patient-semantic contextualization, and a conditioned hypernetwork, the framework effectively resolves task interference, adapts to patient heterogeneity, and mitigates cross-center representation shifts.

2. Methods

Fig. 2 illustrates the proposed unified architecture. To support clinical CCM assessment and regression objectives, we first construct clinically induced visual relations and apply task-specific feature filtering to decouple the visual evidence required by different objectives. The selected visual evidence is then contextualized with task-aware patient semantics. Finally, a shared hypernetwork jointly conditioned on patient semantics and task identity dynamically generates prediction parameters for each patient-task pair.

2.1. Relation-guided task-specific feature extraction

A shared visual encoder must support heterogeneous clinical objectives, but these objectives do not necessarily rely on the same CCM evidence. Moreover, relations learned purely from images may reflect both clinically meaningful phenotypes and nuisance variation introduced by acquisition or cohort differences. We therefore address the problem in two successive stages. First, we organize the overall inter-channel relation structure using a clinically induced relation prior, such that the shared visual representation preferentially preserves relations supported by clinical evidence. Second, because a clinically

regularized feature space may still contain channels that are irrelevant to a particular objective, we further identify a compact task-specific subset through coarse-to-fine statistical filtering. The first stage comprises visual relation representation and clinically induced relation alignment, whereas the second performs task-specific robust feature filtering.

2.1.1. Visual relation representation

We first characterize the relation structure already encoded by the visual backbone before introducing clinical guidance. For the s -th CCM image I_s , the ResNet-50 encoder E_{vis} produces $F_s = E_{\text{vis}}(I_s) \in \mathbb{R}^{C \times H \times W}$, where C denotes the number of feature channels and $H \times W$ the spatial resolution. Let F_s^i denote the activation map of channel i . Different channels may have substantially different activation ranges, and directly comparing their responses would therefore allow high-magnitude channels to dominate the relation estimate. We first apply min-max normalization to each activation map and denote the normalized result by $\bar{F}_s^i$.

Each channel describes the spatial response of one learned visual pattern. Examining channels independently, however, does not reveal whether two patterns tend to respond at similar locations in the image. We therefore model pairwise channel relations according to the spatial similarity of their normalized activation maps. Let $\bar{\mathbf{f}}_s^i \in \mathbb{R}^{HW}$ denote the vectorized form of $\bar{F}_s^i$. For the current image mini-batch $\mathcal{B}_{\text{img}}$, the feature relation modeling block computes

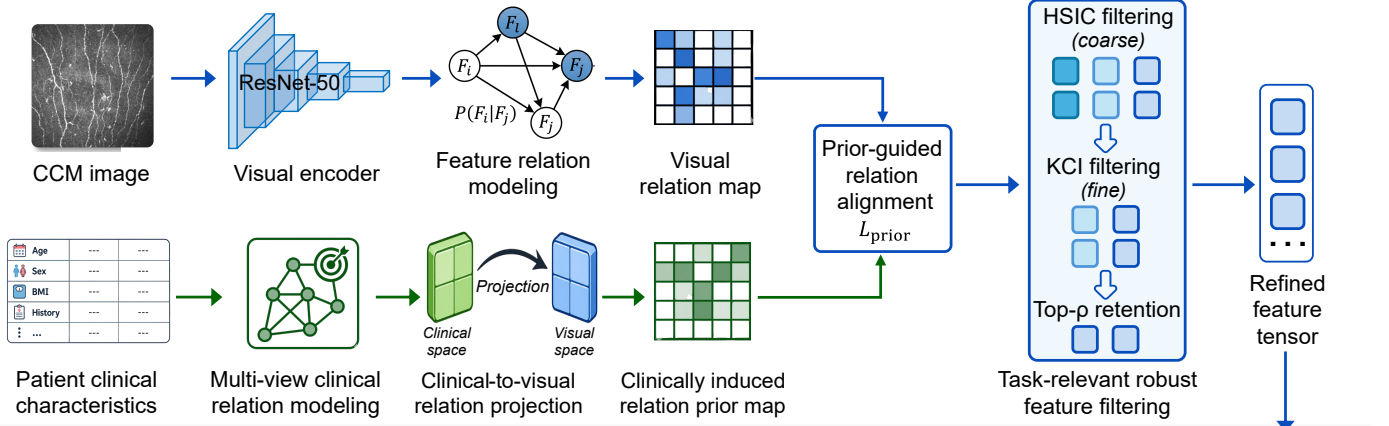
$$M_{\text{vis}}(i, j) = \frac{1}{|\mathcal{B}_{\text{img}}|} \sum_{s \in \mathcal{B}_{\text{img}}} \frac{\langle \bar{\mathbf{f}}_s^i, \bar{\mathbf{f}}_s^j \rangle}{\|\bar{\mathbf{f}}_s^i\|_2 \|\bar{\mathbf{f}}_s^j\|_2 + \epsilon} \quad (1)$$

A large $M_{\text{vis}}(i, j)$ therefore indicates that channels i and j repeatedly activate at similar spatial locations across the images in the mini-batch. The resulting $M_{\text{vis}} \in \mathbb{R}^{C \times C}$ summarizes the inter-channel relation structure currently encoded by the visual backbone. Importantly, M_{vis} only describes relations observed from image features and cannot determine whether a particular channel relation reflects clinically meaningful phenotype structure or an incidental association caused by imaging or cohort variation. We therefore construct an independent relation reference from clinical variables and use it to guide the visual relation space.

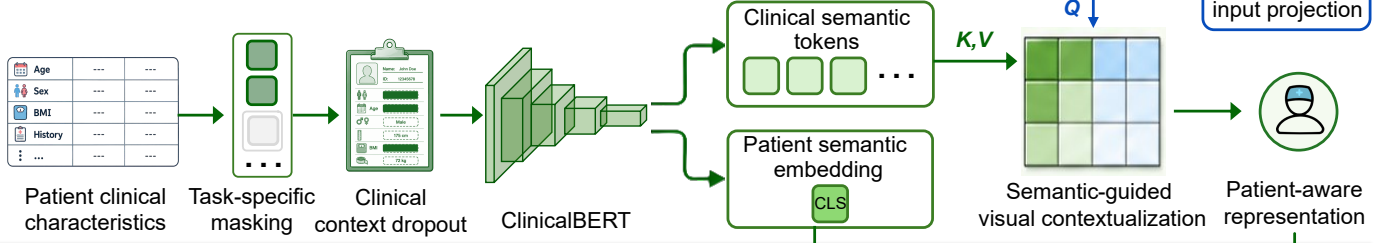
2.1.2. Clinically induced relation alignment

The available clinical variables provide complementary information about disease state, functional measurements, and prognosis. Their pairwise relations can therefore serve as an external reference for organizing visual features. However, clinical variables are heterogeneous, and their relationships may be linear, monotonic, nonlinear, or more general statistical dependencies. A single relation measure is consequently insufficient. Let $C_{\text{clin}} = [c^1, \dots, c^M]$ denote the M clinical supervision variables available in the corresponding training partition. Clinical variables are standardized or encoded according to their data types using statistics derived exclusively from the training partition.

(a) Relation-guided task-specific feature extraction



(b) Patient-semantic-guided feature contextualization



(c) Patient-task-conditioned hypernetwork prediction

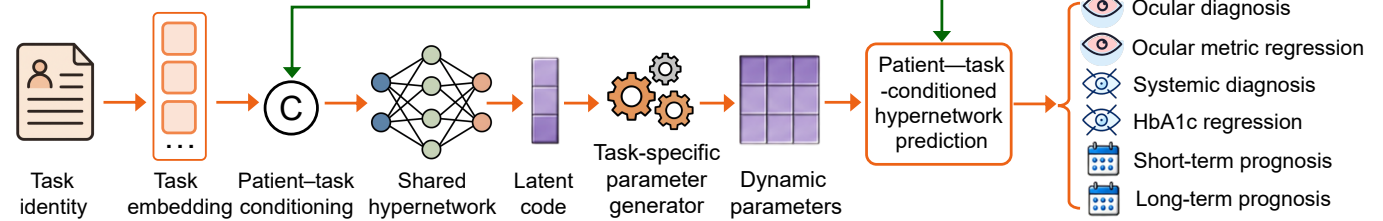


Fig. 2. Overview of the proposed framework. (a) Relation-guided task-specific feature extraction identifies task-relevant CCM evidence under clinical relation guidance. (b) Patient-semantic-guided feature contextualization integrates available patient context with the selected visual evidence. (c) Patient-task-conditioned hypernetwork prediction dynamically adapts the prediction function to each patient-task pair.

We characterize each clinical-variable pair using four complementary relation views: absolute Pearson correlation for linear association, absolute Spearman correlation for monotonic association, distance correlation (dCor) for more general statistical dependence, and normalized mutual information (NMI) for shared information. Pearson and Spearman correlations are used when both variables admit an ordered numerical representation, whereas dCor and NMI accommodate more heterogeneous variable types. For a clinical-variable pair (c^m, c^r) , the applicable relation strengths are collected as

$$\mathbf{a}_{m,r} = [|\rho_P(c^m, c^r)|, |\rho_S(c^m, c^r)|, \text{dCor}(c^m, c^r), \text{NMI}(c^m, c^r)]^T \quad (2)$$

where only statistically applicable views are retained for each variable pair. We use relation magnitude rather than correlation direction because the visual relation map in Eq. (1) represents non-negative similarity strength. The four views need not be equally reliable in the available training population. Rather than averaging them uniformly, we estimate the stability of each view by bootstrap resampling. Let v_k denote the mean bootstrap variance of relation view k over the clinical-variable

pairs for which that view is applicable. A more stable view receives a larger weight, $\alpha_k = \exp(-v_k) / \sum_{\ell} \exp(-v_{\ell})$. For each variable pair, these weights are renormalized over its available views and combined to form the clinical relation matrix $A_{\text{rel}} \in \mathbb{R}^{M \times M}$. Thus, $A_{\text{rel}}(m, r)$ represents the clinically supported relation strength between variables c^m and c^r , while unstable relation estimates contribute less to the final structure.

At this point, A_{rel} and M_{vis} still cannot be compared directly because their nodes have different meanings: A_{rel} is indexed by clinical variables, whereas M_{vis} is indexed by visual channels. We therefore establish a soft correspondence between the two spaces. After the visual encoder has undergone task-supervised warm-up and acquired basic task-discriminative representations, each channel i of sample s is summarized as $z_s^i = \text{GAP}(F_s^i)$. Unlike Eq. (1), which compares two channel maps within the same image, the sequence $\{z_s^i\}_{s=1}^N$ describes how strongly one channel responds across samples, allowing its association with each clinical variable to be estimated.

For each channel-clinical-variable pair (i, m) , we estimate the association strength using the same applicable multi-view relation measures and bootstrap-stability weighting strat-

egy, yielding the channel–clinical association score $R(i, m)$. The relative correspondence is normalized as $\Pi(i, m) = \exp(R(i, m)) / \sum_{r=1}^M \exp(R(i, r))$, and the clinical relation structure is projected into the visual channel space as $M_{\text{prior}} = \Pi A_{\text{rel}} \Pi^T \in \mathbb{R}^{C \times C}$. If two clinical variables exhibit a strong and stable relation, and two visual channels are respectively associated with those variables, the corresponding channel pair receives a stronger relation in M_{prior} . In this way, relations originally defined between clinical variables are converted into a clinically induced relation prior between visual channels. All clinical relation statistics, bootstrap weights, and clinical-to-visual correspondences are estimated exclusively from the training partition, and M_{prior} is fixed once constructed.

The visual relation map M_{vis} and the clinically induced relation prior M_{prior} are now defined over the same C visual channels. Before comparing them, we remove the diagonal self-relations and normalize the remaining structure to eliminate absolute-scale differences:

$$O(M) = M - \text{Diag}(\text{diag}(M)), \quad \widehat{M} = \frac{O(M)}{\|O(M)\|_F + \epsilon} \quad (3)$$

The prior-guided relation alignment objective is then

$$\mathcal{L}_{\text{prior}} = \frac{1}{C(C-1)} \left\| \widehat{M}_{\text{vis}} - \widehat{M}_{\text{prior}} \right\|_F^2 \quad (4)$$

This objective does not force visual features to reproduce individual clinical measurements. Instead, it encourages the relative relation pattern among visual channels to resemble the clinically supported reference structure. Because M_{prior} is fixed after estimation, optimization updates the visual encoder while leaving the clinical reference unchanged. The relation alignment therefore organizes the overall inter-channel feature structure using clinical evidence, but a well-organized shared feature space does not imply that every channel is informative for every clinical objective. We therefore perform a second stage to identify the visual channels most relevant to each task.

2.1.3. Task-specific robust feature filtering

For each task t , task-relevant robust feature filtering follows a coarse-to-fine strategy. HSIC first evaluates the marginal dependence between a visual channel and the task target, whereas KCI evaluates whether this dependence remains after accounting for admissible non-target clinical variables. The final channel budget is then controlled only by top- ρ retention. Because these statistics operate across samples rather than spatial pixels, each channel map is converted into the sample-level descriptor $x_s^i = [\text{GAP}(F_s^i), \text{GMP}(F_s^i)]$, where GAP summarizes the overall response and GMP preserves the strongest localized activation. Let $X_i = \{x_s^i\}_{s=1}^N$ denote the descriptors of channel i over the training samples and Y_t the target representation of task t . For multi-output regression, all target dimensions are treated jointly.

We quantify the marginal dependence between each channel and the task target using the standard Hilbert–Schmidt Independence Criterion (HSIC) (Gretton *et al.*, 2005). No additional proportion-based truncation is introduced at this stage;

all channels evaluated by the HSIC filtering are retained together with their dependence scores as the coarse candidate pool S_t^{HSIC} . Marginal dependence alone is insufficient because a channel may appear target-related through another observed clinical variable. We therefore further evaluate conditional dependence using KCI. Let Z_t denote the admissible non-target clinical variables for task t , excluding the direct prediction target, future measurements, and variables that directly define or reveal that target. For each channel in S_t^{HSIC} , we compute its KCI conditional dependence score with Y_t given Z_t using the standard kernel conditional independence formulation (Zhang *et al.*, 2011). No additional proportion-based truncation is applied to the KCI output; all channels retained by the conditional screening form S_t^{KCI} . The channels supported by both stages are defined as $S_t^{\text{sig}} = S_t^{\text{HSIC}} \cap S_t^{\text{KCI}}$. Thus, HSIC provides marginal target-relevance information, whereas KCI further evaluates whether the channel–target relation remains after accounting for the admissible non-target clinical context.

To ensure a fixed feature budget across tasks, we define $K = \max(1, \lfloor \rho C \rfloor)$ with $0 < \rho \leq 1$ and retain exactly K channels for every task. If $|S_t^{\text{sig}}| \geq K$, channels in S_t^{sig} are ranked by their KCI conditional dependence scores and the top K form S_t . If $|S_t^{\text{sig}}| < K$, all channels in S_t^{sig} are retained and the remaining positions are filled from $S_t^{\text{HSIC}} \setminus S_t^{\text{sig}}$ in descending order of HSIC dependence score. For sample s and task t , the refined feature tensor is therefore $F_{\text{ref},s,t} = F_s[S_t, :, :] \in \mathbb{R}^{K \times H \times W}$. Different tasks may retain different channel combinations, but the feature budget is fixed to K .

At this point, the model has determined *which visual evidence is relevant to each task*. However, the same selected CCM phenotype may still correspond to different clinical states across patients. The next stage therefore introduces patient semantics to determine *how this task-relevant visual evidence should be interpreted for the individual patient*.

2.2. Patient-semantic-guided feature contextualization

Task-specific filtering determines which visual evidence is relevant to each clinical objective, but it does not determine how that evidence should be interpreted for an individual patient. Similar CCM appearances may correspond to different patient states when demographic, systemic, ocular, or symptomatic context differs. We therefore construct a task-safe patient representation from the available clinical information and use it to contextualize the selected visual evidence, yielding a patient-aware representation.

2.2.1. Task-aware patient semantic encoding

Unlike the clinically induced relation prior, which is estimated only from the training partition and fixed thereafter, the patient-semantic branch remains active during inference. It must therefore use only information that would genuinely be available when a particular task is requested. To enforce this requirement and prevent target leakage, we construct a task-safe patient context through *task-specific masking*. For patient s and task t , we denote the available clinical characteristics by C_s^{avail} , which may include demographic information, disease history,

systemic conditions, ocular characteristics, and free-text symptoms. The direct prediction target is denoted by C_t^{target} , variables unavailable at the intended prediction time by C_t^{future} , and variables that directly define or strongly reveal the requested outcome by C_t^{proxy} . The task-safe context is defined inline as $C_{s,t}^{\text{safe}} = C_s^{\text{avail}} \setminus (C_t^{\text{target}} \cup C_t^{\text{future}} \cup C_t^{\text{proxy}})$, where $\setminus$ denotes set difference, i.e., the target, future, and proxy fields are removed from the available patient information. The task-specific admissible field sets are predefined before model training, and potential proxy fields are excluded through clinical review.

Even after task-specific masking, the amount of observed clinical information can vary substantially between patients. We therefore use the *clinical context dropout* block to expose the semantic branch to different levels of context completeness during training. For a patient–task pair with $K_{s,t}$ currently observed task-safe fields, the number of additionally removed fields is sampled as

$$k_{s,t} \sim \text{Uniform}\{0, \dots, K_{s,t}\} \quad (5)$$

after which $k_{s,t}$ fields are selected uniformly without replacement. Thus, $k_{s,t} = 0$ preserves all observed task-safe fields, whereas $k_{s,t} = K_{s,t}$ simulates the complete absence of patient-specific clinical context.

The remaining fields are heterogeneous structured variables and therefore cannot be directly processed by the pretrained language encoder in a unified form. We serialize them into a standardized clinical sentence $S_{s,t}$ using a fixed linguistic template. The template consists of a natural-language scaffold with field placeholders: “A {age}-year-old {sex} patient with systemic status {systemic_status}, medical history {medical_history}, ocular status {ocular_status}, and presenting symptoms {symptom_text}.” The expressions enclosed in braces denote field placeholders and are replaced by the corresponding values during both training and inference. When a field is unavailable or removed by context dropout, its corresponding phrase is omitted from the serialized sentence. Numerical measurements are retained in their original continuous form rather than discretized. The resulting sentence is encoded by *Clinical-BERT* (Alsentzer et al., 2019) as $H_{\text{txt},s,t} = E_{\text{txt}}(S_{s,t}) \in \mathbb{R}^{L \times d_{\text{txt}}}$, where L denotes the token length and d_{txt} the text embedding dimension. The token-level representation $H_{\text{txt},s,t}$ forms the *clinical semantic tokens* used for visual contextualization, whereas the corresponding [CLS] representation $h_{s,t}^{\text{cls}} \in \mathbb{R}^{d_{\text{txt}}}$ forms the *patient semantic embedding* used for subsequent prediction adaptation. When no task-safe clinical field remains after masking and context dropout, the semantic encoder is bypassed and $H_{\text{txt},s,t} = \mathbf{0}$ and $h_{s,t}^{\text{cls}} = \mathbf{0}$.

2.2.2. Patient-semantic-guided visual contextualization

The preceding task-specific feature extraction determines *which* visual evidence is relevant to task t , whereas patient semantics provide the context needed to determine *how* that evidence should be interpreted for patient s . The *semantic-guided visual contextualization* block therefore uses visual tokens as queries and clinical semantic tokens as keys and values, allowing each visual location to retrieve patient context relevant to its interpretation. The selected feature tensor $F_{\text{ref},s,t} \in \mathbb{R}^{K \times H \times W}$ is

rearranged into spatial tokens $X_{\text{vis},s,t} = \text{Flatten}_{\text{spatial}}(F_{\text{ref},s,t})^T \in \mathbb{R}^{HW \times K}$. Although all tasks use the same channel budget K , different tasks retain different channel identities. We therefore use a task-specific input projection $W_{\text{in},t} \in \mathbb{R}^{K \times d_a}$ to map each task-specific channel representation into the shared attention space, yielding $X_{s,t} = X_{\text{vis},s,t} W_{\text{in},t} \in \mathbb{R}^{HW \times d_a}$.

Patient semantics are then injected through residual cross-modal attention:

$$X_{s,t}^{\text{ctx}} = \text{LN}[X_{s,t} + \text{Attn}(X_{s,t} W_Q, H_{\text{txt},s,t} W_K, H_{\text{txt},s,t} W_V)] \quad (6)$$

where $\text{Attn}(\cdot)$ denotes standard scaled dot-product attention, W_Q , W_K , and W_V are learnable projections, and $\text{LN}(\cdot)$ denotes layer normalization. The attention operation preserves the spatial visual tokens while supplementing each token with patient-specific semantic information relevant to that visual evidence. Because the subsequent prediction stage operates on a single patient-level representation, the contextualized tokens are aggregated by average pooling as $f_{s,t} = \text{AvgPool}(X_{s,t}^{\text{ctx}}) \in \mathbb{R}^{d_a}$, yielding the *patient-aware representation*. The residual connection preserves an image-based representation when semantic context is unavailable. The resulting $f_{s,t}$ is patient-aware; the next stage further adapts the representation-to-outcome mapping according to patient semantics and task identity.

2.3. Patient–task-conditioned hypernetwork prediction

The patient-aware representation already integrates task-relevant visual evidence with patient context, but a conventional task-specific head would still apply the same representation-to-outcome mapping to every patient assigned to task t . We therefore adapt not only the representation but also the prediction function itself. Patient semantics and task identity are first combined into a joint condition; a shared hypernetwork transforms this condition into a common latent code, from which a task-specific parameter generator produces the prediction parameters for the requested task. This complements semantic contextualization: the former adapts the representation, whereas the hypernetwork adapts the subsequent mapping.

Patient semantics describe the patient state but do not specify which clinical objective is requested. Each task $t \in \mathcal{T}$ is therefore represented by a learnable task embedding $e_t \in \mathbb{R}^{d_e}$. The framework considers six objectives: ocular neuroimmune diagnosis, systemic neuropathy diagnosis, ocular-surface reference metric regression, HbA1c regression, short-term treatment prognosis, and long-term postoperative prognosis. Because the prediction function depends jointly on patient state and requested objective, the *patient–task conditioning* block forms $p_{s,t} = [e_t \parallel h_{s,t}^{\text{cls}}]$, where $\parallel$ denotes concatenation. The shared hypernetwork $\mathcal{H}$ then maps this condition to a common latent code $g_{s,t} = \mathcal{H}(p_{s,t}) \in \mathbb{R}^{d_h}$, allowing common patient–task conditioning patterns to be shared across objectives.

Different clinical objectives have different output spaces and dimensionalities. A lightweight *task-specific parameter generator* G_t therefore converts the shared latent code into the weight and bias of the dynamic prediction head:

$$[\text{vec}(W_{s,t}); b_{s,t}] = G_t(g_{s,t}) \quad (7)$$

where $W_{s,t} \in \mathbb{R}^{d_a \times o_t}$, $b_{s,t} \in \mathbb{R}^{o_t}$, and o_t denotes the output dimensionality of task t . Thus, patients assigned to the same task can receive different prediction parameters according to their patient semantics. The generated parameters are applied to the patient-aware representation as $\hat{y}_{s,t} = f_{s,t}^T W_{s,t} + b_{s,t}$. When patient-specific clinical information is unavailable, $h_{s,t}^{\text{cls}} = \mathbf{0}$ and the hypernetwork is conditioned by task identity alone through $p_{s,t} = [e_t \parallel \mathbf{0}]$.

2.4. Model optimization and inference

The framework is optimized in two sequential stages. Stage 1 establishes the relation-guided visual encoder and fixed task-specific channel subsets, thereby determining *which* visual evidence is used by each objective. Stage 2 then learns how this evidence is contextualized by patient information and mapped to the requested outcome. This separation prevents the patient-semantic and hypernetwork components from being optimized on an unstable task-specific visual input space.

Stage 1: relation-guided visual representation and task-specific feature selection. Stage 1 begins with task-supervised visual warm-up using $\mathcal{L}_{\text{warm}} = \mathcal{L}_{\text{aux}}$, where cross-entropy is used for classification objectives and mean squared error for regression objectives. After warm-up convergence, the clinically induced relation prior is estimated from the corresponding training partition and fixed. The visual encoder is then further optimized using

$$\mathcal{L}_{\text{rel}} = \mathcal{L}_{\text{aux}} + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}} \quad (8)$$

after which the task-specific channel subsets $\{S_t\}_{t \in \mathcal{T}}$ are determined and fixed for Stage 2.

Stage 2: patient-semantic contextualization and adaptive prediction. With the Stage 1 visual encoder and task-specific channel subsets fixed, ClinicalBERT, the patient-semantic-guided visual contextualization block, the shared hypernetwork, and the task-specific parameter generators are optimized jointly. Clinical context dropout is applied only during training. Because the number of observed samples can differ across tasks, the prediction loss is averaged within each task and then equally across the tasks represented in the current mini-batch. Let $\mathcal{B}_t$ denote the samples of task t in the current mini-batch and $\mathcal{T}_{\mathcal{B}} = \{t : |\mathcal{B}_t| > 0\}$. The task prediction loss is

$$\mathcal{L}_{\text{task}} = \frac{1}{|\mathcal{T}_{\mathcal{B}}|} \sum_{t \in \mathcal{T}_{\mathcal{B}}} \frac{1}{|\mathcal{B}_t|} \sum_{s \in \mathcal{B}_t} \ell_t(\hat{y}_{s,t}, y_{s,t}) \quad (9)$$

where ℓ_t denotes the task-appropriate prediction loss: cross-entropy for multi-class diagnosis, binary cross-entropy for binary prognosis, and mean squared error for continuous regression.

To constrain the magnitude of the dynamically generated prediction parameters, we apply an L_2 penalty. Let $\mathcal{B} = \{(s, t) : s \in \mathcal{B}_t, t \in \mathcal{T}_{\mathcal{B}}\}$ denote the patient-task pairs in the current mini-batch. The regularization term is

$$\mathcal{L}_{\text{hyper}} = \frac{1}{|\mathcal{B}|} \sum_{(s,t) \in \mathcal{B}} (\|W_{s,t}\|_F^2 + \|b_{s,t}\|_2^2) \quad (10)$$

The complete Stage 2 objective is $\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{task}} + \lambda_{\text{hyper}} \mathcal{L}_{\text{hyper}}$.

Fig. 3. Overview of datasets and clinical evaluation tasks.

Inference. During inference, the requested task t determines the fixed channel subset S_t . The CCM image is encoded by the frozen visual encoder and filtered using S_t . Patient information available at prediction time is processed according to the task-safe rules and encoded into patient semantics; the semantic tokens contextualize the selected visual representation, while the patient semantic embedding and task identity condition the shared hypernetwork to generate $W_{s,t}$ and $b_{s,t}$ and produce $\hat{y}_{s,t}$. All relation priors and task-specific channel subsets are fixed after training, so no relation estimation or feature reselection is performed during inference. When patient information is partially or completely unavailable, the residual visual pathway and task identity continue to support prediction within the same framework.

3. Experiments

3.1. Datasets and experimental setup

The study comprised two cross-sectional disease domains and two longitudinal ocular settings, covering diagnosis, clinical reference metric regression, short-term treatment prognosis, and long-term postoperative prognosis. The corresponding cohort composition, participating centers, image and patient numbers, class distributions, and task-specific targets are summarized in Fig. 3. Internal cohorts were used for model development under patient-level cross-validation, whereas independent patients from previously unseen centers were reserved for external evaluation. The ocular-surface reference metric and HbA1c

regression tasks were derived from the same regression dataset and therefore share the same image and patient counts.

The diagnostic framework encompassed both ocular and systemic assessments. The ocular diagnosis task discriminated between healthy controls (HC), dry eye disease (DED), and neuropathic corneal pain (NCP). The systemic diagnosis task modeled disease progression from HC to type 1 diabetes mellitus without diabetic peripheral neuropathy (DPN−) and with DPN (DPN+).

The ocular diagnosis task discriminated HC, DED, and NCP. A separate ocular-surface reference metric regression setting estimated four complementary clinical endpoints—TBUT, CFS, SIT, and OSDI—from CCM images, providing quantitative functional phenotypes in addition to categorical disease discrimination. The systemic diagnosis task modeled the progression from HC to T1DM without DPN (DPN−) and T1DM with DPN (DPN+). HbA1c was additionally estimated as a continuous metabolic reference, allowing corneal neural phenotypes and glycemic burden to be evaluated within the same systemic assessment framework.

Baseline (T_0) CCM and clinical measurements were acquired before treatment, followed by reassessment at one month (1M). Given the baseline CCM image and clinical information available at T_0 , the model directly predicted the absolute one-month values y_{1M} of TBUT, CFS, SIT, and OSDI. Measurements acquired at 1M were used exclusively as prediction targets and were excluded from model inputs. Preoperative baseline (T_0) CCM was used to predict the six-month outcome after FS-LASIK, distinguishing non-persistent dry eye (NPD) from persistent dry eye (PDE). No postoperative information was available to the model at baseline prediction.

Evaluation protocol. All CCM images were acquired using the Heidelberg Retina Tomograph III with the Rostock Cornea Module (HRT-III/RCM) and analyzed at a spatial resolution of 384×384 pixels. External evaluation therefore assessed cross-center generalization under a common imaging platform rather than cross-device generalization. All data partitioning was performed at the patient level: internal cohorts were evaluated using five-fold cross-validation, while external cohorts from independent centers remained fixed and were not involved in model development or hyperparameter selection.

The study was approved by the appropriate institutional ethics committees of the participating centers and by the Institute of Biomedical Engineering, Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences (20240320-0022R). Each center conducted the study under its corresponding local ethical approval or applicable waiver. All images and associated metadata were anonymized before analysis, and all procedures followed the Declaration of Helsinki.

3.2. Implementation details

A ResNet-50 pretrained on ImageNet was used as the visual encoder, and pretrained ClinicalBERT (Tiny) was used for patient-semantic encoding. For multi-view clinical relation modeling, continuous-variable mutual information was estimated using a standard nearest-neighbor estimator, whereas purely categorical relations were estimated from the empirical

joint distribution. HSIC used RBF kernels for continuous descriptors and continuous or multi-output regression targets and a label kernel for categorical targets, with RBF bandwidths determined by the training-partition median heuristic. KCI used the same median-heuristic bandwidth selection together with generalized cross-validation for kernel residualization. Training followed the two-stage procedure described in Sec. 2.4. AdamW was used with an initial learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , cosine annealing to 1×10^{-6} , and batch size 16. Training was limited to 500 epochs with early stopping after 20 epochs without validation improvement. Based on the hyperparameter sensitivity analysis, the only tuned model hyperparameters, λ_{prior} , ρ , and λ_{hyper} , were fixed to 1.0, 30%, and 5×10^{-4} , respectively.

All experiments used strict patient-level partitioning. Internal results are reported as mean $\pm$ standard deviation across the five held-out folds. For external evaluation, the same fixed external cohort was independently evaluated by the five fold-specific models, with mean $\pm$ standard deviation reflecting variability across models. Repeated predictions corresponding to the same case were aggregated before statistical testing. Statistical comparisons used two-sided paired permutation tests for multi-class AUC and regression metrics and two-sided DeLong tests for binary AUC, with Bonferroni correction for multiple comparisons. All normalization statistics, clinical relation quantities, kernel parameters, and task-specific feature-selection quantities were estimated exclusively from the corresponding training partitions. External cohorts remained completely independent during model development and were accessed only after the model configuration and hyperparameters had been fixed.

3.3. Evaluation metrics

Classification performance was evaluated using class-wise accuracy (ACC) and area under the receiver operating characteristic curve (AUC). Class-wise ACC was calculated as the proportion of correctly classified samples within each class, with the average obtained by macro-averaging across classes. For multi-class tasks, AUC was computed in a one-vs-rest manner and macro-averaged across classes. Regression performance was assessed using root mean square error (RMSE) and mean absolute error (MAE). Higher ACC/AUC and lower RMSE/MAE indicate better performance.

3.4. Comparison methods

We compared our framework with eight representative medical multi-task methods capable of jointly performing diagnostic classification and continuous regression: JMMLRC Brand *et al.* (2020), QCNN (Lao *et al.*, 2025), C-former (Mineo *et al.*, 2024), Multi-Cog (Ma *et al.*, 2026), MM-MIL (Huang *et al.*, 2026), MM-DLS (Liu *et al.*, 2026), DenseMoE (Ding *et al.*, 2025), and KGT Zhao *et al.* (2026). All comparison methods were evaluated using the same data partitions and task-specific experimental settings as the proposed framework to ensure a consistent comparison.

Table 1. Quantitative comparison of ocular neuroimmune assessment on the internal and external cohorts. Reported p -values are calculated against Ours (full). Ours (full) uses complete task-safe patient context; Ours (image) uses image-only inference without patient context.

Methods	Diagnosis						Regression					
	Metrics	HC	DED	NCP	Avg.	p <	Metrics	TBUT	CFS	SIT	OSDI	p <
	Internal validation											
JMMLRC	ACC	0.808 _{±0.048}	0.824 _{±0.049}	0.846 _{±0.032}	0.826 _{±0.039}		RMSE	7.247 _{±0.234}	7.104 _{±0.306}	7.481 _{±0.295}	7.168 _{±0.364}	
	AUC	0.848 _{±0.031}	0.862 _{±0.027}	0.887 _{±0.030}	0.866 _{±0.021}	.001	MAE	5.233 _{±0.266}	5.116 _{±0.137}	5.469 _{±0.283}	5.082 _{±0.156}	.001
QCNN	ACC	0.839 _{±0.022}	0.838 _{±0.027}	0.852 _{±0.043}	0.843 _{±0.024}		RMSE	6.716 _{±0.297}	6.381 _{±0.213}	7.126 _{±0.266}	6.549 _{±0.346}	
	AUC	0.879 _{±0.023}	0.903 _{±0.021}	0.883 _{±0.018}	0.888 _{±0.014}	.001	MAE	4.524 _{±0.136}	4.097 _{±0.224}	4.834 _{±0.267}	4.212 _{±0.144}	.001
C-former	ACC	0.826 _{±0.031}	0.831 _{±0.046}	0.885 _{±0.030}	0.847 _{±0.028}		RMSE	5.846 _{±0.267}	5.723 _{±0.155}	6.357 _{±0.336}	6.874 _{±0.221}	
	AUC	0.872 _{±0.027}	0.886 _{±0.022}	0.920 _{±0.019}	0.893 _{±0.021}	.001	MAE	3.414 _{±0.114}	3.187 _{±0.203}	3.923 _{±0.136}	4.136 _{±0.227}	.001
Multi-Cog	ACC	0.852 _{±0.028}	0.854 _{±0.031}	0.868 _{±0.027}	0.858 _{±0.020}		RMSE	6.129 _{±0.184}	5.751 _{±0.276}	6.596 _{±0.254}	6.012 _{±0.333}	
	AUC	0.894 _{±0.021}	0.915 _{±0.013}	0.897 _{±0.022}	0.902 _{±0.014}	.001	MAE	3.867 _{±0.226}	3.443 _{±0.117}	4.114 _{±0.246}	3.576 _{±0.131}	.001
MM-MIL	ACC	0.817 _{±0.043}	0.815 _{±0.052}	0.859 _{±0.022}	0.830 _{±0.031}		RMSE	6.984 _{±0.315}	6.642 _{±0.195}	7.318 _{±0.386}	6.906 _{±0.255}	
	AUC	0.859 _{±0.018}	0.875 _{±0.021}	0.904 _{±0.023}	0.879 _{±0.016}	.001	MAE	4.896 _{±0.153}	4.423 _{±0.233}	5.084 _{±0.176}	4.653 _{±0.245}	.001
MM-DLS	ACC	0.834 _{±0.037}	0.842 _{±0.041}	0.873 _{±0.019}	0.850 _{±0.022}		RMSE	5.857 _{±0.176}	5.594 _{±0.267}	6.371 _{±0.244}	5.786 _{±0.316}	
	AUC	0.887 _{±0.022}	0.896 _{±0.016}	0.891 _{±0.027}	0.891 _{±0.013}	.001	MAE	3.573 _{±0.217}	3.236 _{±0.106}	3.883 _{±0.243}	3.347 _{±0.121}	.001
DenseMoE	ACC	0.844 _{±0.019}	0.856 _{±0.023}	0.873 _{±0.031}	0.858 _{±0.019}		RMSE	5.636 _{±0.264}	5.246 _{±0.144}	7.164 _{±0.327}	7.513 _{±0.216}	
	AUC	0.895 _{±0.018}	0.905 _{±0.019}	0.902 _{±0.012}	0.901 _{±0.013}	.001	MAE	2.987 _{±0.206}	3.064 _{±0.194}	3.756 _{±0.135}	3.127 _{±0.223}	.001
KGT	ACC	0.847 _{±0.030}	0.859 _{±0.031}	0.887 _{±0.017}	0.864 _{±0.018}		RMSE	5.828 _{±0.283}	5.013 _{±0.158}	7.625 _{±0.361}	7.247 _{±0.244}	
	AUC	0.897 _{±0.019}	0.910 _{±0.016}	0.916 _{±0.018}	0.908 _{±0.014}	.001	MAE	2.746 _{±0.118}	3.154 _{±0.202}	3.621 _{±0.141}	2.889 _{±0.232}	.001
Ours (image)	ACC	0.872 _{±0.043}	0.902 _{±0.029}	0.917 _{±0.024}	0.897 _{±0.025}		RMSE	2.438 _{±0.226}	2.285 _{±0.098}	4.173 _{±0.257}	4.526 _{±0.148}	
	AUC	0.917 _{±0.018}	0.931 _{±0.021}	0.942 _{±0.026}	0.930 _{±0.015}	.001	MAE	1.741 _{±0.184}	1.439 _{±0.167}	2.164 _{±0.098}	1.983 _{±0.186}	.001
Ours (full)	ACC	0.901 _{±0.022}	0.914 _{±0.013}	0.940 _{±0.019}	0.918 _{±0.020}		RMSE	1.699 _{±0.197}	1.574 _{±0.083}	3.564 _{±0.234}	3.773 _{±0.122}	
	AUC	0.938 _{±0.018}	0.950 _{±0.011}	0.963 _{±0.027}	0.950 _{±0.015}	–	MAE	1.092 _{±0.064}	0.815 _{±0.146}	1.500 _{±0.077}	1.348 _{±0.163}	–
External validation												
Multi-Cog	ACC	0.781 _{±0.049}	0.826 _{±0.049}	0.813 _{±0.035}	0.807 _{±0.032}		RMSE	7.491 _{±0.265}	6.862 _{±0.342}	7.683 _{±0.385}	7.128 _{±0.426}	
	AUC	0.803 _{±0.031}	0.805 _{±0.025}	0.854 _{±0.027}	0.821 _{±0.023}	.001	MAE	5.274 _{±0.292}	5.148 _{±0.161}	5.345 _{±0.314}	5.267 _{±0.469}	.001
JMMLRC	ACC	0.802 _{±0.030}	0.847 _{±0.032}	0.839 _{±0.036}	0.829 _{±0.024}		RMSE	6.762 _{±0.333}	6.427 _{±0.196}	6.985 _{±0.441}	6.758 _{±0.289}	
	AUC	0.837 _{±0.031}	0.835 _{±0.029}	0.847 _{±0.020}	0.840 _{±0.019}	.001	MAE	4.789 _{±0.160}	4.214 _{±0.231}	4.962 _{±0.185}	4.463 _{±0.253}	.001
MM-DLS	ACC	0.793 _{±0.036}	0.820 _{±0.041}	0.875 _{±0.016}	0.829 _{±0.025}		RMSE	6.038 _{±0.190}	5.619 _{±0.203}	6.542 _{±0.304}	6.953 _{±0.376}	
	AUC	0.825 _{±0.036}	0.841 _{±0.011}	0.868 _{±0.023}	0.845 _{±0.017}	.001	MAE	3.571 _{±0.231}	3.385 _{±0.088}	4.027 _{±0.252}	4.456 _{±0.137}	.001
C-former	ACC	0.819 _{±0.035}	0.838 _{±0.028}	0.862 _{±0.029}	0.840 _{±0.022}		RMSE	6.145 _{±0.319}	5.746 _{±0.178}	6.418 _{±0.423}	6.172 _{±0.268}	
	AUC	0.848 _{±0.025}	0.846 _{±0.027}	0.864 _{±0.014}	0.853 _{±0.017}	.001	MAE	4.121 _{±0.138}	3.529 _{±0.229}	4.293 _{±0.159}	3.762 _{±0.248}	.001
QCNN	ACC	0.784 _{±0.037}	0.811 _{±0.032}	0.828 _{±0.041}	0.808 _{±0.029}		RMSE	7.174 _{±0.241}	6.558 _{±0.324}	7.419 _{±0.353}	6.912 _{±0.422}	
	AUC	0.814 _{±0.033}	0.834 _{±0.029}	0.843 _{±0.031}	0.831 _{±0.021}	.001	MAE	4.962 _{±0.278}	4.685 _{±0.142}	5.118 _{±0.301}	4.921 _{±0.164}	.001
KGT	ACC	0.796 _{±0.043}	0.843 _{±0.031}	0.854 _{±0.024}	0.831 _{±0.025}		RMSE	6.056 _{±0.298}	5.443 _{±0.189}	6.364 _{±0.398}	5.941 _{±0.262}	
	AUC	0.855 _{±0.012}	0.838 _{±0.016}	0.857 _{±0.026}	0.850 _{±0.014}	.001	MAE	3.727 _{±0.141}	3.446 _{±0.211}	3.948 _{±0.164}	3.675 _{±0.232}	.001
DenseMoE	ACC	0.819 _{±0.024}	0.847 _{±0.029}	0.849 _{±0.030}	0.838 _{±0.020}		RMSE	5.858 _{±0.182}	4.894 _{±0.271}	7.305 _{±0.261}	7.642 _{±0.364}	
	AUC	0.833 _{±0.028}	0.842 _{±0.023}	0.865 _{±0.010}	0.847 _{±0.013}	.001	MAE	3.126 _{±0.222}	3.392 _{±0.100}	3.381 _{±0.243}	3.052 _{±0.124}	.001
MM-MIL	ACC	0.816 _{±0.027}	0.852 _{±0.034}	0.860 _{±0.020}	0.843 _{±0.021}		RMSE	5.971 _{±0.204}	4.682 _{±0.208}	7.593 _{±0.282}	7.424 _{±0.401}	
	AUC	0.853 _{±0.027}	0.855 _{±0.014}	0.869 _{±0.021}	0.859 _{±0.013}	.001	MAE	2.869 _{±0.232}	3.041 _{±0.115}	2.858 _{±0.251}	3.393 _{±0.136}	.001
Ours (image)	ACC	0.856 _{±0.023}	0.888 _{±0.016}	0.893 _{±0.019}	0.879 _{±0.021}		RMSE	2.486 _{±0.219}	1.872 _{±0.083}	4.219 _{±0.244}	4.683 _{±0.161}	
	AUC	0.872 _{±0.025}	0.866 _{±0.006}	0.894 _{±0.011}	0.877 _{±0.020}	.001	MAE	1.815 _{±0.088}	1.364 _{±0.162}	1.937 _{±0.086}	1.834 _{±0.171}	.001
Ours (full)	ACC	0.871 _{±0.028}	0.906 _{±0.024}	0.922 _{±0.031}	0.900 _{±0.022}		RMSE	1.810 _{±0.102}	1.145 _{±0.057}	3.428 _{±0.128}	3.958 _{±0.243}	
	AUC	0.898 _{±0.032}	0.899 _{±0.029}	0.920 _{±0.016}	0.906 _{±0.016}	–	MAE	1.232 _{±0.173}	0.829 _{±0.043}	1.206 _{±0.168}	1.117 _{±0.054}	–

4. Experimental Results

4.1. Quantitative comparison

Tables 1–3 compare the proposed framework with representative multi-task methods across six clinical objectives on both internal and independent external cohorts. Overall, Ours achieved the best performance across diagnosis, continuous regression, and longitudinal prognosis, with the advantage consistently preserved under external validation. Unless otherwise specified, reported p -values are calculated against Ours (full).

For ocular neuroimmune diagnosis, Ours improved the average AUC over the strongest competing method by 0.042 internally (0.950 vs. 0.908) and 0.047 externally (0.906 vs. 0.859). The advantage was particularly pronounced for continuous prediction: ocular-surface reference metric regression MAE decreased from 3.103 to 1.189 internally and from 3.040 to 1.096 externally, while HbA1c regression decreased from 2.571 to 0.820 and from 3.092 to 0.965, respectively. Similar improvements were observed for longitudinal prediction, with long-term prognosis AUC increasing by 0.044 on both the internal (0.973 vs. 0.929) and external (0.916 vs. 0.872) cohorts; Tables 1–3 provide the complete task-wise results.

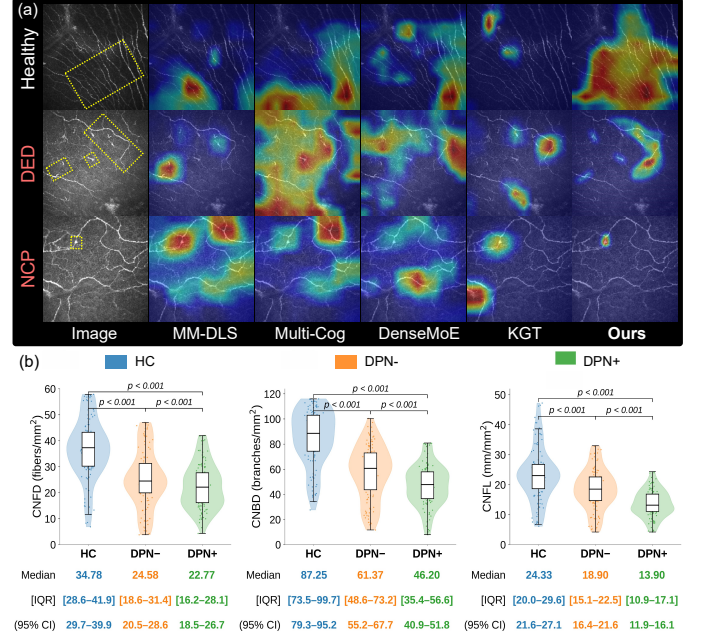
The consistent margins across categorical, continuous, and longitudinal outcomes indicate that the proposed framework provides stable multi-task modeling rather than gains confined

Table 2. Quantitative comparison of systemic neuropathy assessment on the internal and external cohorts.

Methods	Metrics	Diagnosis				Regression			
		HC	DPN-	DPN+	Avg.	p <	Metrics	HbA1c	p <
Internal validation									
JMMLRC	ACC	0.814 $\pm$ 0.041	0.783 $\pm$ 0.037	0.846 $\pm$ 0.034	0.814 $\pm$ 0.028		RMSE	7.286 $\pm$ 0.246	
	AUC	0.858 $\pm$ 0.013	0.834 $\pm$ 0.028	0.872 $\pm$ 0.011	0.855 $\pm$ 0.014	.001	MAE	5.274 $\pm$ 0.259	.001
QCNN	ACC	0.840 $\pm$ 0.033	0.797 $\pm$ 0.032	0.858 $\pm$ 0.030	0.832 $\pm$ 0.022		RMSE	6.733 $\pm$ 0.317	
	AUC	0.880 $\pm$ 0.019	0.851 $\pm$ 0.024	0.887 $\pm$ 0.008	0.873 $\pm$ 0.013	.001	MAE	4.614 $\pm$ 0.136	.001
C-former	ACC	0.831 $\pm$ 0.028	0.825 $\pm$ 0.034	0.863 $\pm$ 0.021	0.840 $\pm$ 0.022		RMSE	5.974 $\pm$ 0.192	
	AUC	0.872 $\pm$ 0.022	0.884 $\pm$ 0.006	0.880 $\pm$ 0.022	0.879 $\pm$ 0.013	.001	MAE	3.453 $\pm$ 0.213	.001
Multi-Cog	ACC	0.853 $\pm$ 0.018	0.806 $\pm$ 0.038	0.874 $\pm$ 0.024	0.844 $\pm$ 0.021		RMSE	6.117 $\pm$ 0.306	
	AUC	0.892 $\pm$ 0.016	0.864 $\pm$ 0.010	0.899 $\pm$ 0.007	0.885 $\pm$ 0.009	.001	MAE	3.896 $\pm$ 0.127	.001
MM-MIL	ACC	0.821 $\pm$ 0.039	0.790 $\pm$ 0.034	0.852 $\pm$ 0.033	0.821 $\pm$ 0.026		RMSE	6.976 $\pm$ 0.231	
	AUC	0.867 $\pm$ 0.011	0.845 $\pm$ 0.024	0.881 $\pm$ 0.010	0.864 $\pm$ 0.012	.001	MAE	4.947 $\pm$ 0.247	.001
MM-DLS	ACC	0.837 $\pm$ 0.025	0.814 $\pm$ 0.036	0.867 $\pm$ 0.018	0.839 $\pm$ 0.021		RMSE	5.914 $\pm$ 0.299	
	AUC	0.884 $\pm$ 0.018	0.860 $\pm$ 0.011	0.893 $\pm$ 0.019	0.879 $\pm$ 0.013	.001	MAE	3.586 $\pm$ 0.122	.001
DenseMoE	ACC	0.851 $\pm$ 0.026	0.809 $\pm$ 0.024	0.872 $\pm$ 0.023	0.844 $\pm$ 0.016		RMSE	4.996 $\pm$ 0.193	
	AUC	0.888 $\pm$ 0.005	0.867 $\pm$ 0.018	0.895 $\pm$ 0.006	0.883 $\pm$ 0.008	.001	MAE	2.853 $\pm$ 0.214	.001
KGT	ACC	0.857 $\pm$ 0.018	0.821 $\pm$ 0.032	0.883 $\pm$ 0.024	0.854 $\pm$ 0.019		RMSE	4.832 $\pm$ 0.285	
	AUC	0.890 $\pm$ 0.016	0.888 $\pm$ 0.005	0.897 $\pm$ 0.018	0.892 $\pm$ 0.005	.001	MAE	2.571 $\pm$ 0.109	.001
Ours (image)	ACC	0.902 $\pm$ 0.028	0.850 $\pm$ 0.023	0.921 $\pm$ 0.032	0.891 $\pm$ 0.022		RMSE	2.148 $\pm$ 0.196	
	AUC	0.931 $\pm$ 0.016	0.904 $\pm$ 0.024	0.934 $\pm$ 0.019	0.923 $\pm$ 0.013	.001	MAE	1.439 $\pm$ 0.154	.001
Ours (full)	ACC	0.912 $\pm$ 0.020	0.882 $\pm$ 0.034	0.934 $\pm$ 0.016	0.909 $\pm$ 0.018		RMSE	1.276 $\pm$ 0.072	
	AUC	0.938 $\pm$ 0.021	0.929 $\pm$ 0.008	0.946 $\pm$ 0.022	0.938 $\pm$ 0.013	–	MAE	0.820 $\pm$ 0.137	–
External validation									
C-former	ACC	0.795 $\pm$ 0.045	0.790 $\pm$ 0.037	0.826 $\pm$ 0.038	0.804 $\pm$ 0.030		RMSE	7.428 $\pm$ 0.259	
	AUC	0.844 $\pm$ 0.030	0.814 $\pm$ 0.028	0.818 $\pm$ 0.027	0.825 $\pm$ 0.022	.001	MAE	5.286 $\pm$ 0.279	.001
MM-MIL	ACC	0.835 $\pm$ 0.027	0.818 $\pm$ 0.035	0.821 $\pm$ 0.029	0.825 $\pm$ 0.024		RMSE	6.784 $\pm$ 0.339	
	AUC	0.850 $\pm$ 0.034	0.831 $\pm$ 0.017	0.843 $\pm$ 0.029	0.841 $\pm$ 0.019	.001	MAE	4.815 $\pm$ 0.448	.001
QCNN	ACC	0.856 $\pm$ 0.041	0.789 $\pm$ 0.027	0.832 $\pm$ 0.027	0.826 $\pm$ 0.025		RMSE	5.823 $\pm$ 0.308	
	AUC	0.862 $\pm$ 0.025	0.850 $\pm$ 0.015	0.859 $\pm$ 0.024	0.857 $\pm$ 0.014	.001	MAE	3.662 $\pm$ 0.135	.001
KGT	ACC	0.814 $\pm$ 0.034	0.806 $\pm$ 0.041	0.870 $\pm$ 0.024	0.830 $\pm$ 0.023		RMSE	6.191 $\pm$ 0.230	
	AUC	0.871 $\pm$ 0.021	0.846 $\pm$ 0.020	0.875 $\pm$ 0.019	0.864 $\pm$ 0.016	.001	MAE	3.979 $\pm$ 0.236	.001
JMMLRC	ACC	0.805 $\pm$ 0.042	0.799 $\pm$ 0.038	0.830 $\pm$ 0.025	0.811 $\pm$ 0.028		RMSE	7.126 $\pm$ 0.345	
	AUC	0.852 $\pm$ 0.027	0.820 $\pm$ 0.024	0.822 $\pm$ 0.038	0.831 $\pm$ 0.022	.001	MAE	5.038 $\pm$ 0.168	.001
DenseMoE	ACC	0.843 $\pm$ 0.035	0.822 $\pm$ 0.020	0.839 $\pm$ 0.026	0.835 $\pm$ 0.021		RMSE	5.989 $\pm$ 0.224	
	AUC	0.857 $\pm$ 0.021	0.839 $\pm$ 0.026	0.855 $\pm$ 0.026	0.850 $\pm$ 0.017	.001	MAE	3.662 $\pm$ 0.244	.001
MM-DLS	ACC	0.831 $\pm$ 0.031	0.820 $\pm$ 0.025	0.843 $\pm$ 0.030	0.831 $\pm$ 0.021		RMSE	5.374 $\pm$ 0.301	
	AUC	0.869 $\pm$ 0.016	0.848 $\pm$ 0.018	0.857 $\pm$ 0.013	0.858 $\pm$ 0.012	.001	MAE	3.147 $\pm$ 0.126	.001
Multi-Cog	ACC	0.842 $\pm$ 0.026	0.825 $\pm$ 0.029	0.860 $\pm$ 0.017	0.842 $\pm$ 0.019		RMSE	4.949 $\pm$ 0.199	
	AUC	0.880 $\pm$ 0.024	0.853 $\pm$ 0.019	0.871 $\pm$ 0.022	0.868 $\pm$ 0.010	.001	MAE	3.092 $\pm$ 0.225	.001
Ours (image)	ACC	0.890 $\pm$ 0.021	0.855 $\pm$ 0.034	0.894 $\pm$ 0.020	0.880 $\pm$ 0.018		RMSE	2.046 $\pm$ 0.197	
	AUC	0.901 $\pm$ 0.029	0.883 $\pm$ 0.017	0.892 $\pm$ 0.026	0.892 $\pm$ 0.019	.001	MAE	1.618 $\pm$ 0.058	.001
Ours (full)	ACC	0.907 $\pm$ 0.026	0.876 $\pm$ 0.032	0.912 $\pm$ 0.019	0.898 $\pm$ 0.020		RMSE	1.385 $\pm$ 0.082	
	AUC	0.923 $\pm$ 0.031	0.902 $\pm$ 0.014	0.915 $\pm$ 0.028	0.913 $\pm$ 0.016	–	MAE	0.965 $\pm$ 0.142	–

Table 3. Quantitative comparison of short-term treatment prognosis and long-term postoperative prognosis on the internal and external cohorts.

Methods	Short-term treatment prognosis						Long-term postoperative prognosis			
	Metrics	TBUT _{IM}	CFS _{IM}	SIT _{IM}	OSDI _{IM}	p <	Metrics	Non-Pers.	Pers.	Avg. p <
<i>Internal validation</i>										
C-former	RMSE	7.383 _{±0.266}	7.123 _{±0.333}	7.534 _{±0.357}	7.282 _{±0.404}		ACC	0.834 _{±0.040}	0.806 _{±0.041}	0.820 _{±0.042}
	MAE	5.269 _{±0.274}	5.114 _{±0.145}	5.408 _{±0.302}	5.029 _{±0.186}	.001	AUC	0.893 _{±0.016}	0.893 _{±0.016}	0.893 _{±0.016}
MM-DLS	RMSE	6.876 _{±0.241}	6.565 _{±0.314}	7.168 _{±0.312}	6.726 _{±0.389}		ACC	0.852 _{±0.034}	0.817 _{±0.037}	0.835 _{±0.038}
	MAE	4.691 _{±0.248}	4.248 _{±0.136}	4.936 _{±0.283}	4.366 _{±0.165}	.001	AUC	0.916 _{±0.019}	0.916 _{±0.019}	0.916 _{±0.019}
JMMLRC	RMSE	6.315 _{±0.305}	6.107 _{±0.196}	6.814 _{±0.392}	7.263 _{±0.268}		ACC	0.843 _{±0.029}	0.846 _{±0.035}	0.845 _{±0.023}
	MAE	3.639 _{±0.124}	3.386 _{±0.213}	4.183 _{±0.167}	4.419 _{±0.244}	.001	AUC	0.908 _{±0.023}	0.908 _{±0.023}	0.908 _{±0.023}
DenseMoE	RMSE	6.364 _{±0.232}	6.044 _{±0.309}	6.649 _{±0.298}	6.201 _{±0.370}		ACC	0.865 _{±0.030}	0.824 _{±0.034}	0.845 _{±0.036}
	MAE	4.085 _{±0.237}	3.631 _{±0.126}	4.319 _{±0.273}	3.758 _{±0.156}	.001	AUC	0.923 _{±0.007}	0.923 _{±0.007}	0.923 _{±0.007}
QCNN	RMSE	7.107 _{±0.256}	6.890 _{±0.328}	7.395 _{±0.333}	6.954 _{±0.401}		ACC	0.827 _{±0.044}	0.819 _{±0.036}	0.823 _{±0.039}
	MAE	4.934 _{±0.261}	4.571 _{±0.133}	5.177 _{±0.295}	4.697 _{±0.168}	.001	AUC	0.901 _{±0.012}	0.901 _{±0.012}	0.901 _{±0.012}
Multi-Cog	RMSE	6.143 _{±0.319}	5.932 _{±0.201}	6.436 _{±0.406}	6.093 _{±0.277}		ACC	0.846 _{±0.028}	0.837 _{±0.039}	0.842 _{±0.024}
	MAE	3.764 _{±0.131}	3.310 _{±0.226}	4.011 _{±0.164}	3.530 _{±0.248}	.001	AUC	0.912 _{±0.021}	0.912 _{±0.021}	0.912 _{±0.021}
KGT	RMSE	6.217 _{±0.297}	5.505 _{±0.181}	6.903 _{±0.361}	7.364 _{±0.246}		ACC	0.866 _{±0.026}	0.834 _{±0.028}	0.850 _{±0.029}
	MAE	3.167 _{±0.115}	3.400 _{±0.208}	3.805 _{±0.142}	3.024 _{±0.223}	.001	AUC	0.923 _{±0.013}	0.923 _{±0.013}	0.923 _{±0.013}
MM-MIL	RMSE	6.124 _{±0.308}	5.808 _{±0.185}	7.132 _{±0.269}	6.867 _{±0.354}		ACC	0.871 _{±0.019}	0.843 _{±0.033}	0.857 _{±0.017}
	MAE	2.847 _{±0.121}	3.239 _{±0.206}	4.085 _{±0.244}	2.724 _{±0.135}	.001	AUC	0.929 _{±0.014}	0.929 _{±0.014}	0.929 _{±0.014}
Ours (image)	RMSE	3.247 _{±0.229}	3.186 _{±0.112}	4.392 _{±0.278}	4.639 _{±0.164}		ACC	0.904 _{±0.021}	0.880 _{±0.036}	0.892 _{±0.040}
	MAE	1.938 _{±0.084}	1.625 _{±0.168}	2.483 _{±0.102}	1.812 _{±0.174}	.001	AUC	0.958 _{±0.028}	0.958 _{±0.028}	0.958 _{±0.028}
Ours (full)	RMSE	2.472 _{±0.112}	2.352 _{±0.196}	3.643 _{±0.163}	3.714 _{±0.248}		ACC	0.923 _{±0.032}	0.904 _{±0.026}	0.914 _{±0.030}
	MAE	1.199 _{±0.167}	1.043 _{±0.045}	1.827 _{±0.184}	1.166 _{±0.056}	–	AUC	0.973 _{±0.015}	0.973 _{±0.015}	0.973 _{±0.015}
<i>External validation</i>										
C-former	RMSE	7.476 _{±0.395}	7.260 _{±0.262}	7.617 _{±0.466}	7.387 _{±0.324}		ACC	0.884 _{±0.023}	0.758 _{±0.057}	0.821 _{±0.028}
	MAE	5.416 _{±0.188}	5.175 _{±0.269}	5.360 _{±0.203}	5.091 _{±0.297}	.001	AUC	0.826 _{±0.042}	0.826 _{±0.042}	0.826 _{±0.042}
DenseMoE	RMSE	6.913 _{±0.359}	6.605 _{±0.224}	7.156 _{±0.443}	6.736 _{±0.294}		ACC	0.876 _{±0.037}	0.816 _{±0.032}	0.846 _{±0.035}
	MAE	4.753 _{±0.164}	4.313 _{±0.241}	4.814 _{±0.178}	4.434 _{±0.266}	.001	AUC	0.853 _{±0.018}	0.853 _{±0.018}	0.853 _{±0.018}
MM-MIL	RMSE	6.165 _{±0.220}	5.857 _{±0.298}	6.705 _{±0.297}	7.183 _{±0.373}		ACC	0.918 _{±0.015}	0.786 _{±0.048}	0.852 _{±0.019}
	MAE	3.680 _{±0.223}	3.436 _{±0.128}	4.139 _{±0.254}	4.369 _{±0.147}	.001	AUC	0.857 _{±0.030}	0.857 _{±0.030}	0.857 _{±0.030}
KGT	RMSE	6.408 _{±0.337}	6.092 _{±0.212}	6.646 _{±0.411}	6.219 _{±0.285}		ACC	0.895 _{±0.031}	0.808 _{±0.030}	0.852 _{±0.030}
	MAE	4.126 _{±0.155}	3.685 _{±0.234}	4.391 _{±0.170}	3.803 _{±0.258}	.001	AUC	0.864 _{±0.016}	0.864 _{±0.016}	0.864 _{±0.016}
MM-DLS	RMSE	7.157 _{±0.263}	6.846 _{±0.335}	7.390 _{±0.351}	6.978 _{±0.406}		ACC	0.889 _{±0.030}	0.769 _{±0.045}	0.829 _{±0.026}
	MAE	4.989 _{±0.276}	4.648 _{±0.159}	5.035 _{±0.298}	4.752 _{±0.172}	.001	AUC	0.835 _{±0.029}	0.835 _{±0.029}	0.835 _{±0.029}
JMMLRC	RMSE	6.194 _{±0.230}	5.882 _{±0.307}	6.429 _{±0.311}	6.015 _{±0.376}		ACC	0.912 _{±0.024}	0.783 _{±0.039}	0.848 _{±0.019}
	MAE	3.817 _{±0.238}	3.374 _{±0.127}	4.066 _{±0.271}	3.587 _{±0.143}	.001	AUC	0.869 _{±0.012}	0.869 _{±0.012}	0.869 _{±0.012}
Multi-Cog	RMSE	6.069 _{±0.314}	5.549 _{±0.181}	7.004 _{±0.378}	7.472 _{±0.258}		ACC	0.901 _{±0.016}	0.812 _{±0.036}	0.857 _{±0.017}
	MAE	3.494 _{±0.132}	3.651 _{±0.214}	3.541 _{±0.147}	3.070 _{±0.229}	.001	AUC	0.866 _{±0.020}	0.866 _{±0.020}	0.866 _{±0.020}
QCNN	RMSE	6.783 _{±0.216}	6.047 _{±0.302}	6.826 _{±0.271}	8.239 _{±0.403}		ACC	0.923 _{±0.021}	0.814 _{±0.027}	0.869 _{±0.024}
	MAE	3.349 _{±0.223}	3.971 _{±0.119}	3.283 _{±0.203}	2.802 _{±0.117}	.001	AUC	0.872 _{±0.013}	0.872 _{±0.013}	0.872 _{±0.013}
Ours (image)	RMSE	3.315 _{±0.231}	3.436 _{±0.124}	4.127 _{±0.263}	4.839 _{±0.184}		ACC	0.957 _{±0.026}	0.841 _{±0.027}	0.899 _{±0.033}
	MAE	2.584 _{±0.096}	2.315 _{±0.188}	1.762 _{±0.069}	1.815 _{±0.165}	.001	AUC	0.898 _{±0.019}	0.898 _{±0.019}	0.898 _{±0.019}
Ours (full)	RMSE	2.583 _{±0.115}	2.678 _{±0.206}	3.418 _{±0.146}	3.979 _{±0.265}		ACC	0.974 _{±0.029}	0.865 _{±0.031}	0.920 _{±0.021}
	MAE	1.805 _{±0.177}	1.770 _{±0.069}	1.076 _{±0.143}	1.176 _{±0.047}	–	AUC	0.916 _{±0.017}	0.916 _{±0.017}	0.916 _{±0.017}

**Fig. 4. Qualitative comparison and neuro-morphometric analysis. (a) Grad-CAM visualizations across representative CCM phenotypes. (b) Distributions of CNFD, CNBD, and CNFL across HC, DPN-, and DPN+.**

4.3. Ablation study

We conducted progressive ablations across six clinical objectives to examine the contribution of each innovation and its key components (Fig. 5). Within each module, components are introduced sequentially on top of the preceding configuration, while the macro-level setting progressively integrates the three modules from the visual backbone to the complete framework. This design exposes the marginal contribution of each step rather than only comparing the final model with a single baseline. Overall, performance generally improves as the components are progressively introduced across diagnosis, regression, and prognosis.

Starting from the visual backbone (V0), clinically induced relation alignment, HSIC filtering, and KCI filtering with top- ρ retention are added progressively to form V1–V3. The most pronounced gain appears in ocular regression, where MAE decreases from 2.153 to 1.699. This improvement is consistent with the role of relation-guided task-specific feature extraction: clinically induced relations first regularize acquisition- and cohort-sensitive feature dependencies, while subsequent HSIC/KCI filtering retains factors that remain informative for the specific task. Fig. 5 further shows consistent improvements across the remaining objectives, supporting the effectiveness of task-specific robust feature filtering.

T0–T3 progressively introduce task-aware patient semantic encoding, semantic-guided visual contextualization, and the residual visual pathway on top of the task-specific visual representation. A representative effect is observed in systemic regression, where MAE decreases from 1.502 to 1.188. The gain indicates that patient semantics provide complementary context for interpreting visually similar CCM patterns, thereby reducing ambiguity that cannot be resolved from image features alone. Consistent trends across the other tasks are shown in

in abnormal cases, where competing methods often produce broader or fragmented activations, whereas our model concentrates more strongly on localized nerve alterations. This suggests that the learned representation relies more on disease-related structural evidence rather than non-specific image appearance.

Fig. 4(b) further evaluates clinically relevant corneal nerve loss in systemic neuropathy. Nerves from the held-out internal test sets and complete external cohort were segmented using CCM-Pro (Li et al., 2025b), from which CNFD, CNBD, and CNFL were quantified following established clinical definitions (Alam et al., 2017). Group differences were assessed using the Kruskal–Wallis test with Bonferroni-corrected pairwise comparisons; all three metrics decreased progressively from HC to DPN- and DPN+ ($p < 0.001$). As a representative example, median CNFL decreases from 24.33 to 18.90 and 13.90 mm/mm², respectively. This monotonic pattern is consistent with progressive corneal nerve degeneration in diabetic peripheral neuropathy (Zhang et al., 2025; Gad et al., 2022), providing complementary evidence that the model emphasizes clinically relevant neural phenotypes.

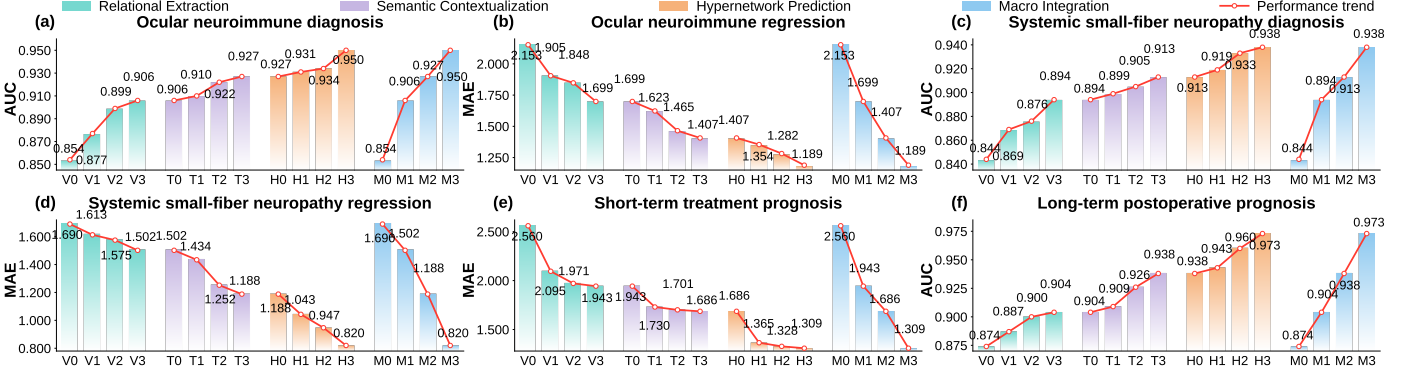


Fig. 5. Progressive ablation analysis of the proposed framework on the internal validation cohort. Component- and module-level contributions across six clinical objectives. Under the progressive construction, $V3=T0$, $T3=H0$, and $H3=M3$.

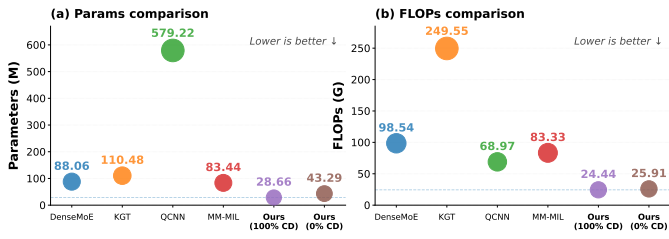


Fig. 6. Computational complexity comparison under Ours (image) and Ours (full) configurations.

Fig. 5.

H0–H3 progressively replace a static prediction head with task embedding, patient–task conditioning, and task-specific parameter generation. The clearest improvement is observed in systemic regression, where MAE decreases from 1.188 to 0.820. This result supports the central role of the hypernetwork: rather than forcing heterogeneous diagnosis, regression, and prognosis objectives to share a fixed prediction function, the model adapts its parameters to both patient context and task identity. Fig. 5 shows that this benefit is consistently preserved across the remaining objectives.

M0–M3 progressively integrate Relation-guided Task-specific Feature Extraction, Patient-semantic-guided Feature Contextualization, and Patient–task-conditioned Hypernetwork Prediction. Performance progressively improves as the modules are integrated; for example, ocular diagnosis AUC increases from 0.854 to 0.950 from M0 to M3. This macro-level progression indicates that the three modules address complementary sources of difficulty—visual nuisance variation, patient-level clinical heterogeneity, and task-level prediction heterogeneity—rather than providing redundant improvements. Together, the progressive trends in Fig. 5 validate the complete design from robust task-specific feature construction to patient-aware contextualization and adaptive prediction.

4.4. Computational complexity analysis

As shown in Fig. 6, the proposed framework maintains low model complexity while achieving superior predictive performance. Under the complete-context setting, Ours (full) uses 43.29M parameters, substantially fewer than representative baselines such as MM-MIL (83.44M) and KGT (110.48M),

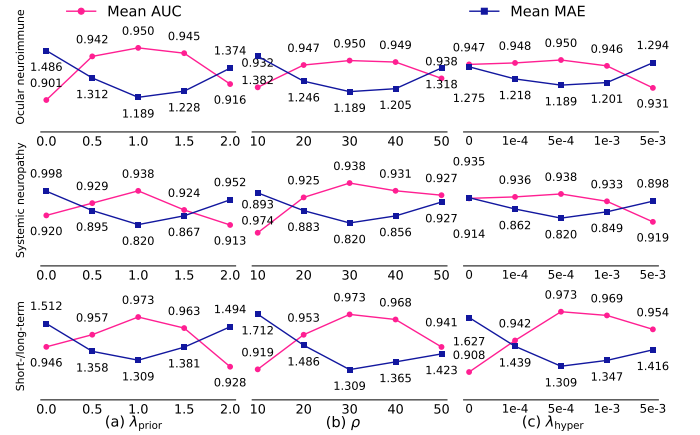


Fig. 7. Hyperparameter sensitivity across clinical tasks on the internal validation cohort. Effects of λ_{prior} , ρ , and λ_{hyper} on ocular neuroimmune, systemic neuropathy, and short-/long-term performance.

while requiring only 25.91G FLOPs. Under image-only inference, Ours (image) removes the ClinicalBERT-based clinical-context branch from the active pathway, reducing the active inference complexity to 28.66M parameters and 24.44G FLOPs. These results indicate that the performance gains are not driven by model-scale expansion. Instead, the framework relies on shared representations and lightweight patient–task-conditioned parameter generation rather than separate large task-specific networks, yielding a favorable performance–complexity trade-off and efficient computation.

4.5. Hyperparameter sensitivity analysis

The best overall configuration was obtained with $\lambda_{\text{prior}} = 1.0$, $\rho = 30\%$, and $\lambda_{\text{hyper}} = 5 \times 10^{-4}$ (Fig. 7). Across all task groups, performance generally improved toward intermediate settings and declined when the corresponding constraint became excessive. Specifically, moderate λ_{prior} balances relational guidance and visual flexibility, $\rho = 30\%$ preserves informative task-specific factors while suppressing redundant channels, and moderate λ_{hyper} provides sufficient regularization without limiting patient–task-adaptive prediction. These trends indicate a stable trade-off among relation alignment, feature selection, and hypernetwork adaptation.

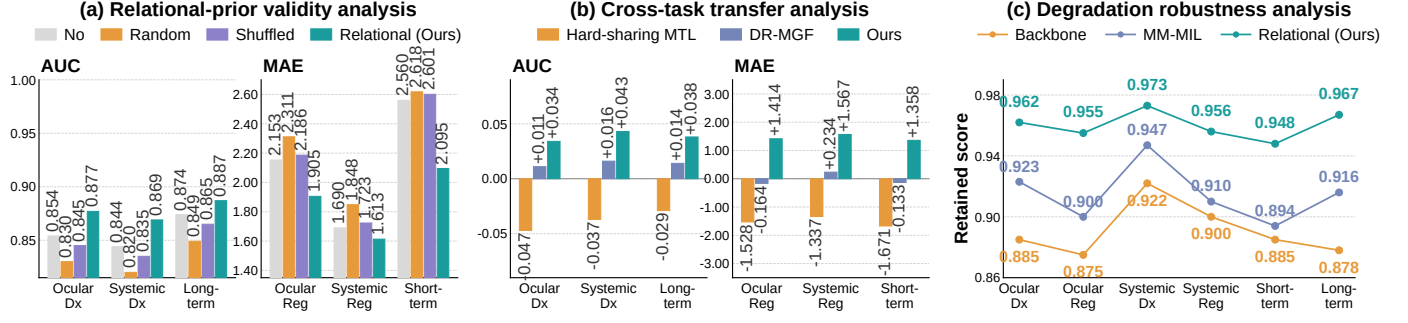


Fig. 8. Mechanism and robustness analyses. (a) Validity of the clinically induced relation structure across diagnosis, regression, and prognosis tasks. (b) Cross-task transfer relative to single-task learning. (c) Performance retention under input degradation for the backbone, MM-MIL, and relational feature extraction.

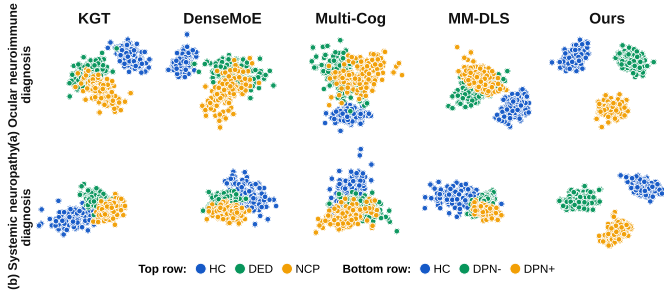


Fig. 9. t-SNE visualization of learned representations. Top: ocular neuroimmune diagnosis; bottom: systemic neuropathy diagnosis.

5. Discussion and Further Analysis

Beyond overall predictive performance, we further examined why the proposed framework is effective and whether its learned outputs remain clinically meaningful. Specifically, the discussion focuses on three complementary aspects: mechanism and robustness analyses to explain the contribution of the core design under task and acquisition variability, representation visualization to assess whether disease-related features become more discriminative, and clinical validity analyses to determine whether the predicted reference metrics and prognostic outputs preserve established disease-related patterns and longitudinal relevance. Together, these analyses connect model behavior with both methodological effectiveness and clinical plausibility.

5.1. Mechanism and robustness analyses

Fig. 8 examines three properties underlying the effectiveness of the proposed framework: whether clinically induced relations provide meaningful rather than generic regularization, whether patient-task adaptation reduces interference among heterogeneous objectives, and whether relation-guided feature extraction improves resistance to acquisition degradation.

Fig. 8(a) tests whether the gain from relation-guided feature extraction originates from meaningful clinical structure. The *No* setting removes relation alignment, whereas *Random* replaces the clinical relation structure with an independently sampled symmetric matrix and *Shuffled* disrupts variable correspondence while preserving its value distribution; *Relational (Ours)* uses the clinically induced relations defined in Sec. 2.1.1. The clearest effect is observed for short-term prognosis, where MAE

decreases from 2.560 without relation guidance to 2.095 with the clinically induced structure, while both random and shuffled relations fail to reproduce this improvement. Fig. 8(a) shows the same tendency across the remaining objectives. This contrast indicates that the benefit is not explained by imposing an arbitrary relational constraint; rather, preserving clinically supported dependencies provides a more informative organization of the latent visual space.

Fig. 8(b) evaluates whether the proposed patient-task-conditioned prediction mechanism alleviates interference among diagnosis, regression, and prognosis objectives. Cross-task transfer is defined as $\Delta = \text{Method} - \text{Single}$ for AUC-based tasks and $\Delta = \text{Single} - \text{Method}$ for MAE-based tasks, where *Method* denotes each evaluated multi-task approach and positive values indicate improvement. Hard-sharing multi-task learning produces negative transfer across all six tasks, whereas the gradient-conflict-oriented DR-MGF (Sun et al., 2026) baseline alleviates part of this interference but does not consistently yield positive transfer. In contrast, our framework improves all six objectives relative to their single-task counterparts; for example, systemic regression shows a gain of +1.567. This consistent transfer supports the proposed design in which task-specific feature selection limits inappropriate representation sharing and patient-task conditioning further adapts the prediction function to heterogeneous clinical objectives.

Fig. 8(c) further tests whether relation-guided feature extraction stabilizes representations under simulated acquisition degradation, including blur, noise, illumination variation, and contrast perturbation. Performance retention is defined as $R = M_{\text{deg}}/M_{\text{clean}}$ for AUC-based tasks and $R = M_{\text{clean}}/M_{\text{deg}}$ for MAE-based tasks, such that higher values consistently indicate better preservation. A representative example is ocular regression, for which the retained score increases from 0.875 with the backbone and 0.900 with MM-MIL to 0.955 with *Relational (Ours)*. The same ordering is observed across the six objectives in Fig. 8(c). These results are consistent with the intended role of clinically induced relation alignment: constraining feature dependencies toward stable clinical structure reduces reliance on degradation-sensitive visual correlations and improves representation stability under suboptimal imaging conditions.

Together, these analyses provide complementary mechanistic evidence for the framework: clinically induced relations organize the visual representation in a meaningful manner, patient-

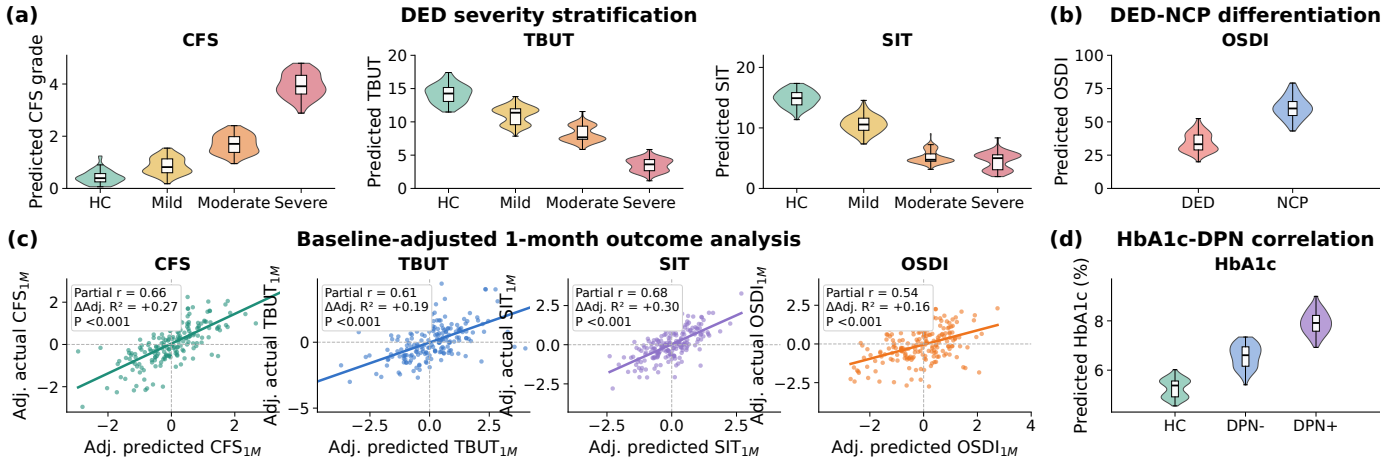


Fig. 10. Clinical validity of metric regression and prognosis. Associations with disease severity, phenotype differentiation, longitudinal outcomes, and systemic metabolic status are evaluated.

task adaptation mitigates heterogeneous-task interference, and relation-guided feature extraction improves robustness to acquisition degradation.

5.2. *t*-SNE visualization

Fig. 9 examines whether the proposed framework improves the organization of disease-related representations rather than only the final prediction scores. Across both ocular neuroimmune and systemic neuropathy diagnosis, Ours produces more compact intra-class clusters and clearer inter-class separation than KGT, DenseMoE, Multi-Cog, and MM-DLS. The difference is particularly evident for clinically adjacent phenotypes: DED and NCP remain substantially overlapped for several competing methods, whereas the proposed framework separates HC, DED, and NCP more distinctly; a similar reduction in overlap is observed among HC, DPN-, and DPN+. This representation-level separation is consistent with the intended role of task-specific feature extraction and patient-semantic contextualization, suggesting that the performance gains arise from more discriminative phenotype representations rather than solely from changes in the prediction head.

5.3. Clinical validity of metric regression and prognosis

Beyond numerical regression accuracy, we examined whether the predicted reference metrics reproduce clinically expected disease patterns (Fig. 10).

As shown in Fig. 10 (a), for DED severity, predicted CFS increased progressively from HC to severe DED, whereas TBUT and SIT decreased, consistent with established associations between ocular-surface signs and increasing DED severity (Sullivan *et al.*, 2010; Begley *et al.*, 2003).

As shown in Fig. 10 (b), predicted OSDI also remained substantially higher in NCP than in conventional DED, consistent with the greater symptom burden reported in neuropathic corneal pain (Chin *et al.*, 2024). Importantly, these severity and phenotype labels were independently established by unanimous consensus of three senior clinicians, and were not used to construct the corresponding predictions. The observed trends

therefore provide an independent assessment of clinical concordance.

As shown in Fig. 10 (c), we further evaluated whether the predicted one-month outcomes retained information beyond the corresponding baseline measurements. Image-level predictions from the same patient were first averaged to obtain a single patient-level estimate. For each endpoint, the observed one-month value y_{1M} was then modeled using both the actual baseline value y_0 and the held-out prediction $\hat{y}_{1M}$, i.e., $y_{1M} = \beta_0 + \beta_1 y_0 + \beta_2 \hat{y}_{1M} + \epsilon$. Partial correlations were computed after adjustment for y_0 , and the increase in adjusted R^2 over the baseline-only model quantified the additional explanatory value of the prediction. As shown in the figure, all four endpoints, including CFS, TBUT, SIT, and OSDI, remained significantly associated with their observed one-month outcomes after baseline adjustment, indicating that the model captured longitudinal information beyond baseline status.

As shown in Fig. 10 (d), predicted HbA1c increased from HC to DPN- and DPN+, consistent with the established association between glycemic burden and diabetic peripheral neuropathy (Braffett *et al.*, 2020; Pinto *et al.*, 2020). Collectively, these findings suggest that the predicted reference metrics preserve clinically plausible ordering, phenotype differences, and longitudinal relevance.

6. Conclusion

We addressed the challenge of unified CCM-based assessment across ocular neuroimmune and systemic neuropathy diagnosis, continuous clinical metric regression, and longitudinal prognosis. The proposed framework combines relation-guided task-specific feature extraction, patient-semantic-guided feature contextualization, and patient-task-conditioned hypernetwork prediction to learn robust task-relevant representations while adapting prediction to individual patients and heterogeneous clinical objectives. Across internal and independent external cohorts, the framework achieved consistently strong performance across categorical, continuous, and prognostic tasks, while the predicted clinical metrics preserved clinically mean-

ingful disease-related patterns. These findings support the potential of CCM as a unified source of imaging-derived biomarkers for disease assessment, quantitative clinical characterization, and longitudinal prediction.

A key limitation is that the clinical objectives were evaluated in task-specific cohorts rather than in a single cohort undergoing the complete set of assessments. Future work should therefore focus on prospective validation in unified, more heterogeneous patient cohorts to further establish generalizability and clinical utility.

7. Acknowledgement

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